

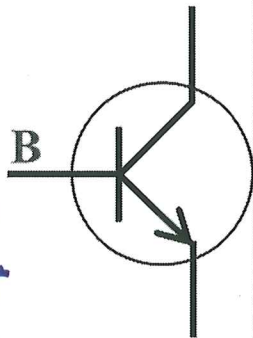
L#12: Transistors

2017-08-16

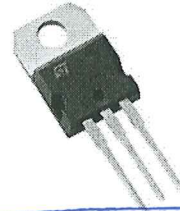
Collector = C

Base

Emitter = E

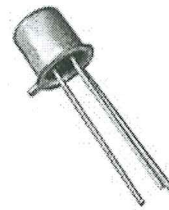


Field effect
Transistor

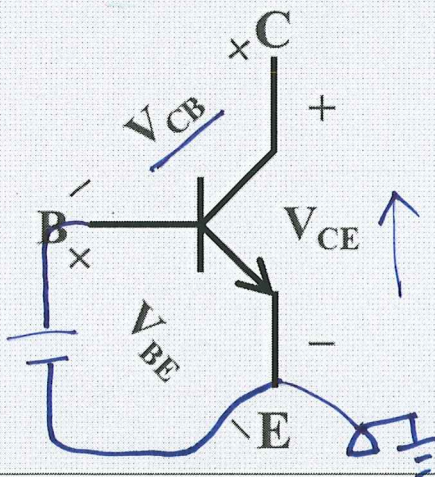


FET

Bipolar Junction
Transistor



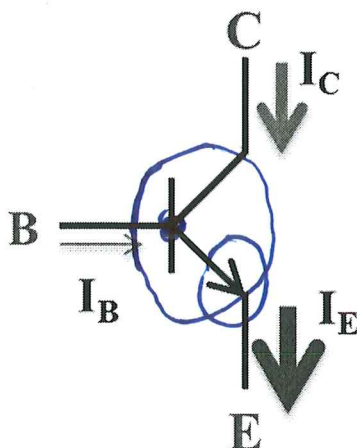
BJT



Voltage:

$$V_{CE} = V_{BE} + V_{CB} = 0.7 + V_{CB}$$

$$V_{BE} = 0.7 \text{ V}$$



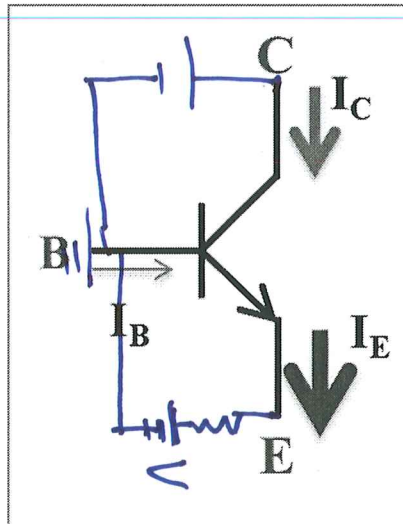
Current:

$$I_E = I_B + I_C$$

$$I_E = I_B + I_C \quad (1)$$

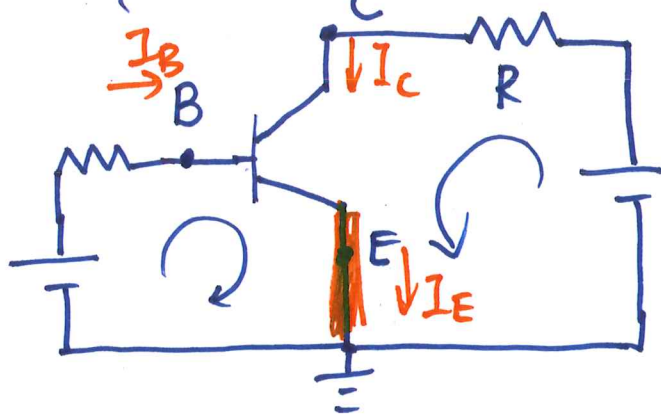
$$I_C = \beta I_B \quad (2)$$

$$I_C = \alpha I_E \quad (3)$$



β : base-emitter current gain (50-100) α : base-emitter current gain (0.98-0.999)

β : common-emitter current gain



$$I_C = \beta I_B \quad (2)$$

1A 0.1A

β : 50-100

α : common-base current gain

$$I_C = \alpha I_E \quad (3), \quad \alpha \rightarrow 1, \quad 0.98-0.99$$

$$\textcircled{1} \Rightarrow \underline{I_E} = I_B + \beta I_B = \underline{(1 + \beta) I_B}$$

$$\textcircled{3} \Rightarrow \cancel{\beta I_B} = \alpha (1 + \beta) \cancel{I_B}$$

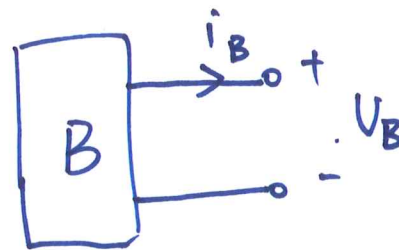
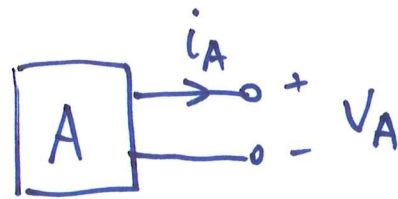
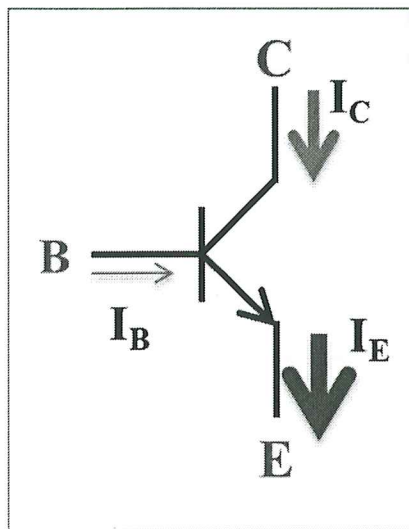
$$\alpha = \frac{\beta}{1 + \beta} \quad \textcircled{A} \quad \beta \text{ is a big number} \Rightarrow \alpha \rightarrow 1, \alpha < 1$$

$$\beta = \alpha + \alpha \beta \Rightarrow (1 - \alpha) \beta = \alpha$$

$$\beta = \frac{\alpha}{1 - \alpha} \quad \textcircled{B}$$

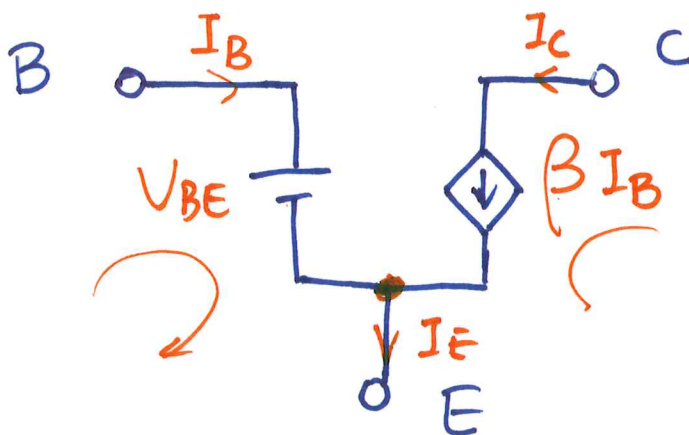
99% $\downarrow$ 0.99

Equivalent network model



$$i_A = i_B \quad \& \quad V_A = V_B$$

$\Rightarrow$ A is equivalent to B



- $V_{BE} = 0.7V$.

- $I_C = \beta I_B$

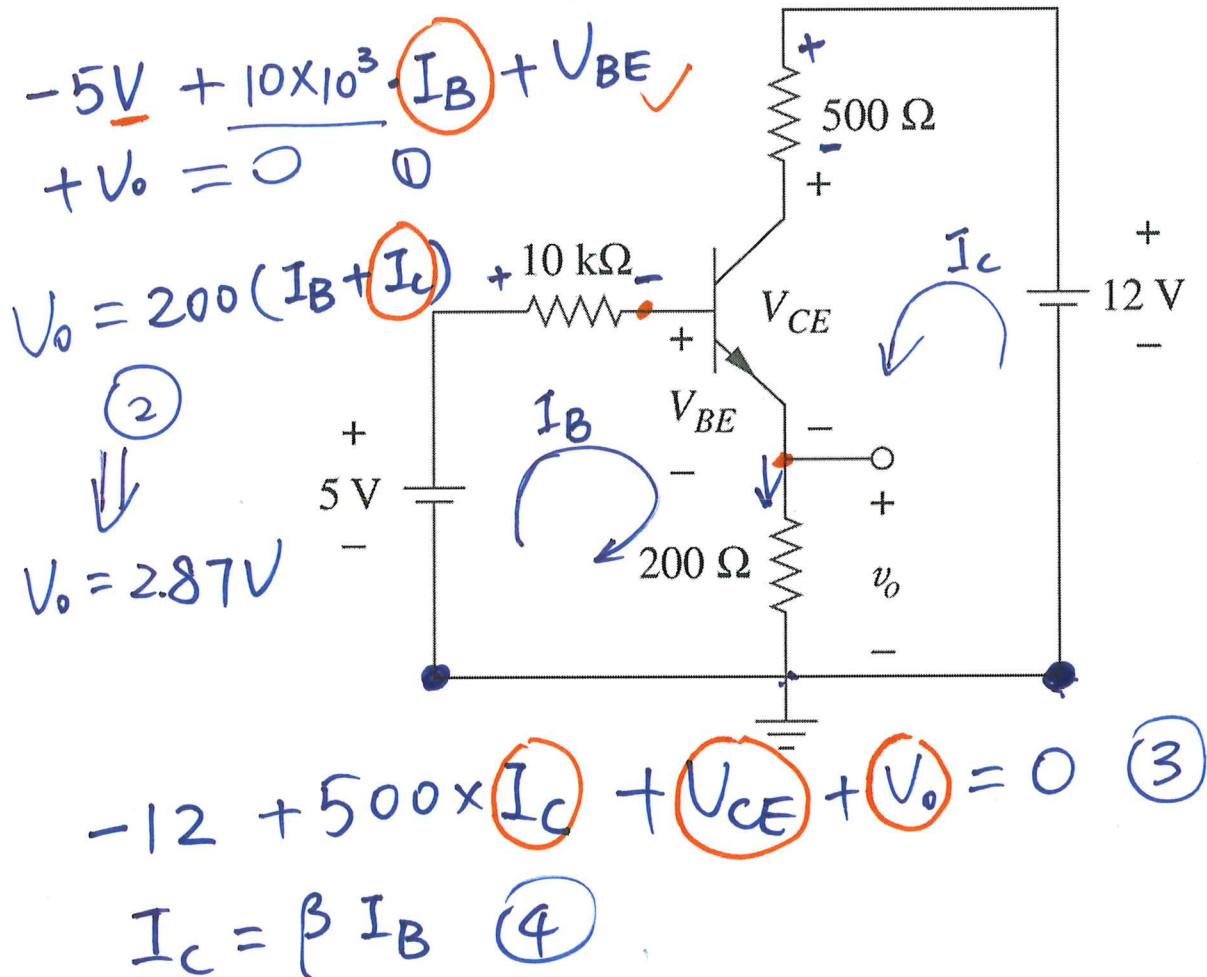
- $I_E = I_B + I_C$

Transistor circuit analysis

- Mesh analysis
- Nodal analysis - equivalent network model

Example #1:

Determine v_o and V_{CE} for the transistor circuit in which $V_{BE} = 0.7\text{ V}$ and $\beta = 100$.



$$-5 + 10^4 I_B + 0.7 + 200(I_B + \beta I_B) = 0$$

$$10^4 I_B + 200 \times 101 I_B = 4.3$$

$$\begin{array}{r} 20000 \\ 200 \\ \hline 10000 \end{array}$$

$$30200 I_B = 4.3$$

$$\Rightarrow I_B = 0.1424 \text{ mA}$$



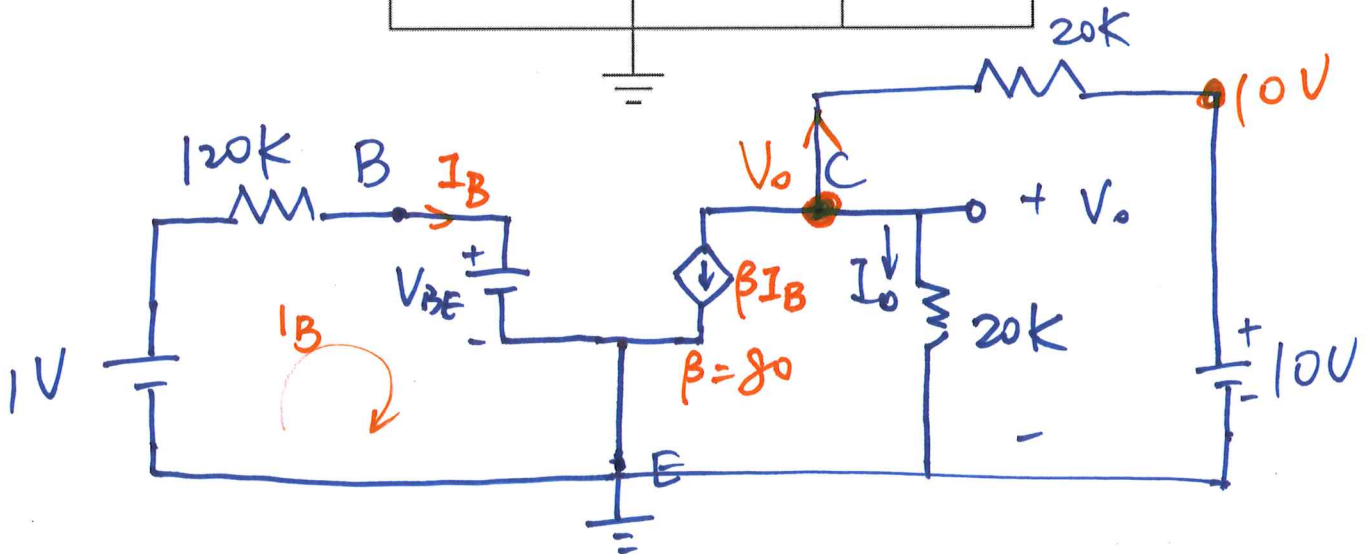
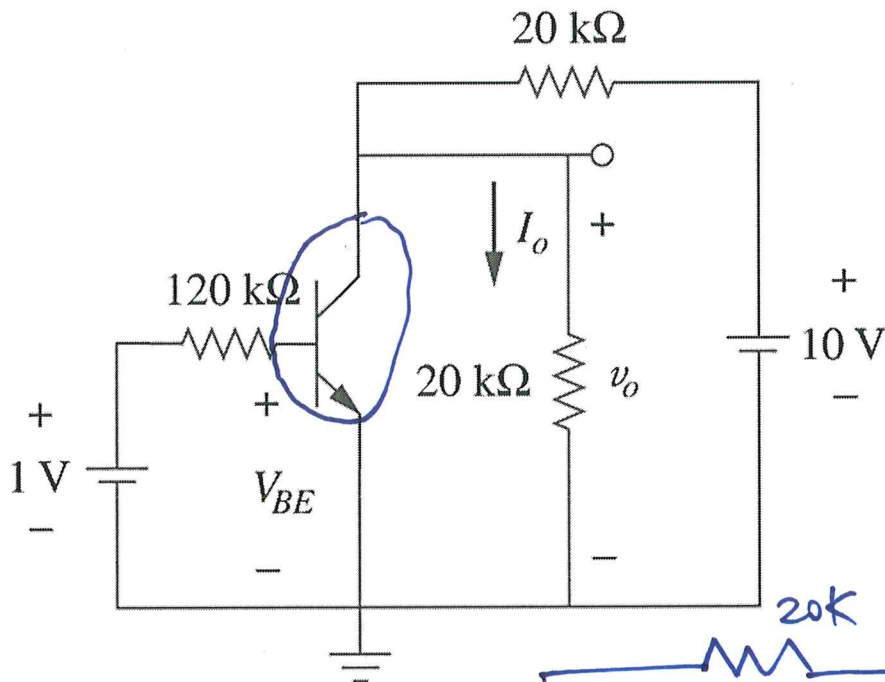
$$-12 + 500 \times 100 \times 0.1424 \times 10^{-3} + V_{CE}$$

$$+ 200(I_B + 100I_B) = 0$$

$$V_{CE} = 2V.$$

Example #2:

Determine v_o and I_o for the transistor circuit in which $V_{BE} = 0.7\text{ V}$ and $\beta = 80$.



$$-1 + 120 \times 10^3 \times I_B + 0.7 = 0 \Rightarrow I_B = 2.5 \mu\text{A}$$

$$\beta I_B + \frac{V_o}{20\text{k}} + \frac{V_o - 10}{20\text{k}} = 0$$

$$80 \times 2.5 \times 10^{-6} \times 20 \times 10^3 + 2V_o = 10$$

$$200$$

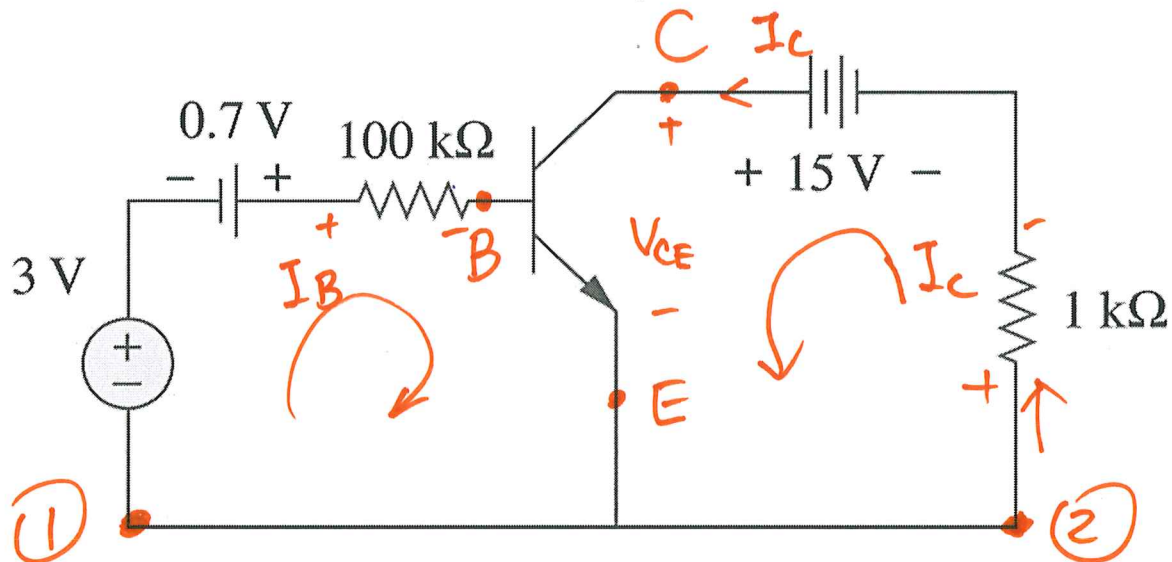
$$4 + 2V_o = 10$$

$$\Rightarrow V_o = 3\text{ V}$$

$$I_o = \frac{3\text{ V}}{20\text{ k}\Omega} = 150 \mu\text{A}$$

Example #3:

Determine V_{CE} and I_B for the transistor circuit in which $V_{BE} = 0.7 \text{ V}$ and $\beta = 100$.



$$\underbrace{-3}_{(1)} - \underbrace{0.7}_{V_{BE} = 0.7} + \underbrace{100 \times 10^3 \times I_B}_{100 \text{ k}\Omega \times I_B} + \underbrace{V_{BE}}_{=0.7} = 0 \quad (1)$$

$$I_B = 30 \mu\text{A}$$

$$10^3 \times I_C - 15 + V_{CE} = 0 \quad (2)$$

$$10^3 \times \underbrace{100}_{\beta} \times \underbrace{30 \times 10^{-6}}_{I_B} - 15 = -V_{CE}$$

$$V_{CE} = 12 \text{ V}$$